DARSHAN LAXMAN GONEPPANAVAR

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SUMMARY

Software engineer with 5 years building, owning, and operating production systems across full stack web applications, distributed backend services, payments, data pipelines, and applied AI and ML. Designs RESTful and GraphQL microservices and event driven services on AWS (Lambda, API Gateway, SQS, SNS, S3, EC2, CloudWatch) in Python, TypeScript, JavaScript, and Node.js, with React and Next.js front ends. Shipped natural language search and a retrieval augmented chatbot with LLM output evaluation, and trains and optimizes transformer models in PyTorch with custom GPU kernels, distributed training, and quantized inference. Sole owner of a platform migration serving 10,000 users, integrator of PayPal and RazorPay payment workflows with idempotent, fault tolerant webhook processing, and mentor to junior engineers. Completing an MS in Computer Science in December 2026.

EDUCATION

California State University, Long Beach — **Long Beach, CA**

MS in Computer Science | GPA: 3.7/4.0 — *January 2025 to December 2026*

- Relevant Coursework: Data Structures and Algorithms, Distributed Computing, Artificial Intelligence, Machine Vision, Computer Architecture, Blockchain

Visvesvaraya Technological University — **Bengaluru, KA, India**

Bachelor of Engineering in Computer Science — *August 2017 to August 2021*

EXPERIENCE

California State University, Long Beach — **Long Beach, CA**

Full Stack Developer, Graduate Assistant — *March 2025 to Present*

- Achieved complete transaction capture on the PayPal NCP vendor migration by architecting a three layer verification pipeline across PDT, REST API, and webhook channels with SHA256-RSA signature validation, OAuth2, and idempotent handlers covering retry, replay, and partial failure on AWS API Gateway and Lambda, with CloudWatch monitoring inbound event delivery.
- Maintained 99.9 percent stability for 10,000 users as sole developer owning an end to end migration of a legacy enterprise platform to PHP 8, making all architecture, data modeling, and rollout decisions independently and remediating security vulnerabilities including SQL injection, XSS, and access control gaps.
- Reduced staff record lookup time by 60 percent by building a natural language search experience over 62 database tables, designing the query interface, result presentation, and empty and loading states on a Python retrieval layer using OpenAI embeddings and ChromaDB.
- Resolved 70 percent of routine staff inquiries without human intervention by building a retrieval augmented chatbot on FastAPI and GPT-4, tuning chunking and retrieval strategies and evaluating LLM output correctness and data sensitivity before release.
- Replaced recurring manual financial reporting with a filterable reporting interface backed by SQL aggregation, date range controls, and one click Excel export using SheetJS.
- Established the project's first knowledge base across 17 Confluence pages and set up JIRA Kanban workflows for tracking work across releases.

Hutech Solutions PVT LTD — **Bengaluru, KA, India**

Software Developer (early stage startup, six person team) — *February 2024 to January 2025*

- Sustained over 2,000 requests per second and cut latency 17 percent by building Node.js and TypeScript middleware with Redis caching against PostgreSQL, writing queries, migrations, and schema changes across releases.
- Processed over 50,000 weekly transactions and raised transaction success rate 16 percent by implementing a RazorPay payment integration with AWS SNS and SQS webhook processing, exponential backoff, and idempotent handlers.
- Reduced vulnerability surface 90 percent by hardening all user facing endpoints with input validation, XSS protection, and parameterized queries.
- Secured user access across the platform by building JWT based single sign on and role based access control for login and authorization.
- Built and deployed backend microservices exposing GraphQL and REST APIs on Kubernetes, owning service design through release.
- Built real time data pipelines with Kafka and Spark and analytics dashboards on React, Django REST, and PostgreSQL.
- Increased user engagement 20 percent for 10,000 monthly active users by building responsive React, Next.js, Redux, and Tailwind CSS interfaces from Figma designs, with sortable and filterable data tables and loading and error states, covered by Jest and React Testing Library.
- Cut data retrieval times 40 percent while handling over 5,000 real time updates daily by integrating Google Maps location tracking views.

Accenture Solutions PVT LTD **Bengaluru, KA, India**

Application Developer Analyst *November 2022 to February 2024*

Application Developer Associate *October 2021 to October 2022*

- Delivered production features across a full stack React and Node.js health insurance platform backed by Oracle serving 178 million customers for a Fortune 500 client, contributing to design reviews and code reviews alongside 12 engineers in Agile sprints.
- Loaded 1 to 50 million rows per run daily by building Informatica ETL mappings and workflows end to end against Oracle, tuning slow and failing loads and automating job control with Unix shell scripts.
- Cut deployment time 75 percent, from 3 hours to 45 minutes, by managing automated Jenkins CI/CD release pipelines with unit and integration testing built into the deployment gate.
- Decreased page load times 20 percent and lifted user retention 5 percent by refactoring frontend architecture, Redux state management, and client side routing.
- Reduced JS and CSS transfer size 18 percent over gzip by adding build time Brotli compression to a MERN asset pipeline.
- Cut new hire onboarding time 50 percent by mentoring junior developers through onboarding and pair programming, establishing coding standards, Git workflows, and documentation practices.

IoTracx PVT LTD **Bengaluru, KA, India**

Blockchain Research Associate *October 2019 to April 2020*

- Evaluated scalability and throughput across public and private blockchain networks to inform architecture choices for decentralized applications.
- Researched distributed consensus algorithms, smart contracts, and DApp architectures, comparing their tradeoffs in fault tolerance and performance.

PROJECTS

Transformer Language Model, Training and Inference from Scratch *PyTorch, Triton, DDP*

- Trained a 124M parameter decoder only transformer on 2.4 billion tokens across two A100 GPUs using distributed data parallel training, bf16 mixed precision, and gradient checkpointing.
- Sped up training 1.9x on softmax and 2.3x on LayerNorm over native PyTorch by writing fused GPU kernels in Triton.
- Cut inference memory 68 percent through INT8 quantization and KV caching at under 0.5 percent perplexity loss.

Vision Language Action Model for Form Automation *SmolVLM-2B, LoRA, React, Node*

- Reached 87 percent action accuracy on held out Workday, Greenhouse, and Lever layouts by fine tuning a 2B parameter vision language model with LoRA to map form screenshots and profile data to executable actions.
- Gained 11 points on ambiguous fields by adding chain of thought reasoning before action tokens.
- Shipped it as a React and Express application with MongoDB capturing user corrections as a retraining signal.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript (ES6+), SQL, PHP, HTML, CSS

Frontend: React.js, Next.js, Redux, Tailwind CSS, jQuery, Ajax, HighCharts, AG Grid, Figma

Backend and APIs: Node.js, Express.js, FastAPI, Django REST, Flask, RESTful APIs, GraphQL, WebSocket, Microservices, Redis

AI and ML: PyTorch, JAX (familiar), Triton, CUDA, Transformer Architecture, LoRA Fine Tuning, Distributed Training (DDP), Mixed Precision, INT8 Quantization, KV Caching, Training and Inference Pipelines, RAG Pipelines, Semantic Search, Embeddings, FAISS, ChromaDB, OpenAI API, Anthropic Claude API, LangChain, LLM Evaluation, Natural Language Understanding (NLU)

Data and Databases: PostgreSQL, Oracle, MySQL, MongoDB, Kafka, Spark, Informatica ETL, SQL Query Optimization, Data Modeling

Cloud and DevOps: AWS (Lambda, API Gateway, SQS, SNS, S3, EC2, CloudWatch), GCP, Azure, Docker, Kubernetes, Jenkins CI/CD, GitHub Actions, Git, Linux, Unix Shell Scripting

Security: OAuth2, JWT Single Sign On, Role Based Access Control, SHA256-RSA Signature Verification, Idempotency and Replay Protection, Input Validation, XSS Prevention, Parameterized Queries

Testing: Jest, React Testing Library, Unit Testing, Integration Testing, Test Driven Development

CS Fundamentals: Data Structures and Algorithms, Object Oriented Design, Distributed Systems, System Design, Event Driven Architecture

Practices: Agile, SDLC, Technical Design, Code Review, Design Review, Production Support, Operational Excellence, Technical Documentation, Mentorship

CERTIFICATIONS

Google Associate Cloud Engineer (2023) | Google Cloud Digital Leader (2023) | Microsoft Azure Administrator Associate AZ-104 (2023) | Microsoft Azure Fundamentals AZ-900 (2022)